

Ryan Glusic

AI Agents and HPC Systems

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HPC engineer at NVIDIA working across GPU infrastructure, Linux, Slurm, and distributed training and inference. Building AI-assisted operations around infrastructure tools, persistent memory, and human approval controls, with a foundation in physics, mathematics, and scientific computing.

TECHNICAL STRENGTHS

AI systems: Agent workflows and tool integration; persistent memory and consolidation; human approval controls; Slack interfaces

Infrastructure: Linux; Slurm; GPU/HPC systems; distributed training and inference; Ansible; Prometheus

Scientific computing: Python; NumPy; Matplotlib; AiiDA; Quantum ESPRESSO; OpenPBS; MPI

EXPERIENCE

NVIDIA

Jul 2022 - Present

IT DevOps Engineer

- Develop an internal AI operations assistant for NVHPC, connecting Slurm monitoring and issue alerts with an interactive Slack interface designed for team use.
- Design agent workflows that connect infrastructure tools, persistent memory and consolidation, and administrator approval and denial gates.
- Develop automated failure diagnosis and diagnostic job rerun workflows, both currently in progress.
- Work across GPU infrastructure, Linux, and Slurm for distributed training and inference workloads.

Data in Science Technologies

Associate HPC Engineer

- Supported administration and maintenance of Bright-based HPC clusters for clients across the US.
- Created Ansible roles to deploy and configure Prometheus exporters and Open OnDemand, and documented installation procedures for client services.
- Built a VirtualGL passthrough layer for NVIDIA Titan GPUs and helped configure Elastic Stack ingestion with Elasticsearch, Filebeat, and Kibana.

RESEARCH

University of Wisconsin-Parkside

Jan 2020 - Jun 2021

Undergraduate Research Assistant

- Administered a computational cluster and integrated OpenPBS to schedule MPI workloads.
- Built a Python/AiiDA interface to coordinate Quantum ESPRESSO phonon calculations and MPI execution through the cluster scheduler.
- Analyzed phonon convergence in silicon and aluminum using Python visualization workflows; presented the research at APS and WiSys.

EDUCATION

University of Wisconsin-Parkside

May 2021

Bachelor of Science in Physics and Mathematics